

IN VITRO EXPERIMENTS ON THE FACTORS DETERMINING THE COURSE OF THE OUTGROWING NERVE FIBER

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EIGHTEEN FIGURES

Innumerable times the outgrowth of developing or regenerating nerve fibers has been watched, while taking place, both outside the organism in tissue culture (Harrison, '07, '10 and others) and in the organism (Harrison, '08; Speidel, '32, '33). Yet to mere observation, the mechanism that operates in guiding the outgrowing nerve fibers into a definite pattern, peripheral and intracentral, has not revealed itself. In an attempt to elucidate the guiding principle, highly valuable contributions have been made both by experimental work and by speculation, and the question has been answered in several ways that sounded rather definitive. However, surveying the existing evidence, one realizes that none of the hypothetical solutions offered can be considered as wholly satisfactory, as long as their experimental foundation is so disproportionately small, and undoubtedly the disproportion between facts and speculation in this field is very great indeed.

A few excellent review papers have, in recent years, summarized the matter (Tello, '23; Herrick, '25; Detwiler, '33). Generally speaking, there coexist in the literature three different viewpoints which, at first glance, seem to be incompati-

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ble with each other: the chemical, the electrical and the mechanical. The chemical viewpoint, developed chiefly by Cajal (1893) and Forssman (1898, '00), considers nerve orientation to be a chemotactic effect: specific substances diffusing from localized centers would 'attract' the nerve fibers. The electrical hypothesis, brought forward by Strasser (1892), Kappers ('17), Child ('21) and others, assumes a direct orienting effect of electric potential differences on the nerve fiber. The mechanical viewpoint, advocated by His (1887), Harrison ('10), and some students of nerve regeneration (Dustin, '10) attributes to solid mechanical structures the role of tracks guiding the nerve fiber on its way. There seemed to be some evidence at hand for every one of these conceptions, but no decisive proof for any of them. Since it could hardly be regarded as satisfactory to assume that in one instance mechanical, in another chemical and in a third electrical agents were the guiding principle, unless they all operated in an indirect way by means of some mechanism common to them all, and since the accumulation of new facts obviously is the only promising way of approach toward a more satisfactory conception, further experimentation is needed.

For this purpose, the method of tissue culture offers itself. In fact, it was with such purposes in mind that Harrison originally devised the method (Harrison, '12). Nerve fibers growing *in vitro* are released from the uncontrollable influences of the organism; they, then, can be again subjected to a variety of well-defined physical and chemical factors, in an attempt to reestablish as many as possible of the particular features of organization that have been lost by their previous isolation. By this procedure, different factors which it would be impossible to dissociate in the organism can be tested separately and checked as to their supposed effectiveness. Intricate and complicated situations of the organism are taken apart, in a sense, and are tested, part by part, under the somewhat standard conditions *in vitro*. Apparently, to deal with the intricate conditions under which nerve orientation takes place in the organism, a similar procedure was needed, such

as would allow a separate check of the chemical, electrical and mechanical factors respectively.

The first attempt of that sort was Harrison's test on the mechanical factor; by adding spider-web fibers to a nerve culture the reaction of outgrowing nerve fibers to solid structures was to be investigated (Harrison, '14). The experiments with nerves were not successful, however, as neither in frog nor in chick cultures did the nerve fibers seem to be affected by the spider web. Next, Ingvar ('20), in Harrison's laboratory, tried to test the electric factor. In his extremely condensed report Ingvar states that he obtained definite orientation effects on nerve fibers grown in an electric current between two electrodes. Ingvar's results, well fitting as they seemed, into those speculations that considered electric potential differences as the orienting factor in nerve growth, were widely acknowledged in competent literature, and once more a problem of fundamental importance appeared to be satisfactorily settled. Yet, it was not. Objections could be raised, not to Ingvar's observations, but to their interpretation as electric effects (compare below). So when in 1929 certain experimental results, proving the leading role of mechanical structures in orienting the growth of cells (P. Weiss, '29), once more emphasized the possibility that mechanical structures might be the factor in orienting the growth as of cells, so of nerves, too, the work described in the following pages was planned. Nerve fibers in tissue culture should be subjected, if possible, to all those conditions (chemical, electrical and mechanical) which were suspected or claimed to be of orienting influence on nerve growth.

From the very beginning, the mechanical factor was in the foreground. The reason for this is found in the suggestive results that I had obtained with other cells, results which demonstrated that cell patterns in vitro arise on the basis of corresponding structural patterns in the medium in which the cells move and grow. It had been shown by Harrison ('14) that growing spindle cells move along microscopic threads; by my experiments it was revealed that it does not

even require fibers of microscopic visibility to orient cells on a structural basis. In short the situation is this: The rod-like fibrin micellae of the blood plasma medium tend to aggregate; by oriented mechanical forces, for instance tension, acting on the system in a definite direction, the micellae can be forced to orient and to aggregate along the lines of force (compare W. J. Schmidt, '24). It is such a pattern of oriented micellar aggregates, then, that serves the growing cells as a system of guiding tracks (principle of 'ultrastructural organization'). Both direction and intensity of growth have been demonstrated to depend on the 'ultrastructural organization' of the medium. It may be maintained that such 'ultrastructural organization,' occurring in the 'ground substances' in the embryo, plays a fundamental role in morphogenesis; the matter has recently been summarized in a general way (P. Weiss, '33).

Many facts pointed to the applicability of the principle of 'ultrastructural organization' to the problem of nerve patterns. Therefore, in 1929, I suggested the possibility that micellar orientations in the embryonic ground substances may be the basic mechanism by which nerve fibers are oriented, and that chemical, or electrical, or other factors, inasmuch as they influence nerve orientation, do not affect the nerve fiber directly but act primarily on the ground substance, creating or altering the ultrastructural patterns therein, which the nerve fibers secondarily follow as their guiding system (P. Weiss, '29, p. 549, and '33, p. 335). This conception has two obvious advantages: first, it conceives a uniform mechanism for all the different forms of orienting influences that have been described, and, secondly, it can easily be subjected to critical experimental tests, as is shown in the following pages.

MATERIAL AND METHODS

All experiments were done with brain fragments or spinal ganglia of chick embryos. If pure nerve growth was desired, brain fragments were used; they were taken either from the mid-brain (embryos incubated from 3 to 8 days) or from the

medulla (embryos incubated from 6 to 14 days). When it seemed desirable to have sheath cells² and fibrocytes growing out as well as nerve processes, spinal ganglia were used. These were taken from embryos of from 10 to 15 days' incubation. Blood plasma of adult chickens was used as culture medium. Although the outgrowth of nerve processes is obtained as well with as without the addition of embryonic extract to the culture medium, in most cases a drop of diluted embryonic extract was added to secure rapid coagulation. Particular reference to methods will be made when describing the experiments.

Although many facts could be observed in the living culture, fixation and selective staining were required both for photographic purposes and for the clear distinction between true nerve processes and other fibrillar structures, e.g., filaments of fibrocytes or fibrin fibers in the clot. Methylene blue, although simple in its application (Lazarenko, '31), did not prove satisfactory for this purpose. Neither did various haematoxylin preparations, among them Held's molybdic acid-haematoxylin. It is evident that a sharp selectivity of stain is offered only by silver impregnations. The modification of Cajal's method by Esaki ('29), which in some cases gave very brilliant results, did not prove satisfactory in others, especially where the thickness of the clot was considerable. Another disadvantage of this method is the long duration of the whole procedure (3 to 4 weeks). The method of Bielschowsky as applied to tissue culture by Grigorieff ('31) did not yield consistent results either, the plasma clot quite often becoming dark colored. I tried then to adapt for the application to tissue culture Oscar Schultze's sodium-hydroxide silver method, which is very successful with frozen sections. Though satisfactory results were obtained at times,

²In cultures of spinal ganglia two types of spindle cells can be observed, a plumper and a slenderer type. Cells of the latter differ in many respects from ordinary fibrocytes: They are extremely elongated and thin, and exhibit a striking tendency to arrange in tandem chains. Also it seems that their nuclei stain in methylene blue slightly differently than fibrocyte nuclei do. There is every reason to believe that these cells are sheath cells and they will be referred to as such in this paper.

I shall not outline the procedure, because in many instances the nerves stained only partially. Finally I fell back on the Bielschowsky (da Fano) method and after some essential modifications succeeded in obtaining very beautiful, consistent and reliable specific impregnations.

The procedure is as follows:

Fixation in 10 cc. formol conc. + 20 cc. alcohol 95 per cent + 40 cc. aq. dest. To every 10 cc. of this solution add 1 drop of ammonia. Leave the preparation in this solution for 2 days or more.

After fixation, wash in 50 per cent alcohol for about 3 minutes. Rinse with fresh 50 per cent alcohol. (Water should not be used at all.) Prolonged washing impairs the subsequent impregnation.

After washing, allow the preparations to stand in 10 per cent silver nitrate at room temperature for 2 days. (If the previous washing has not been sufficient, a cloudy precipitation may appear in the clot, which can be redissolved within a few seconds in a strong solution of ammonia; after thoroughly washing in alcohol the silver nitrate may be applied again.) Upon rinsing off the excess silver nitrate, then dipping the preparation for a few seconds in distilled water, an ammoniacal silver nitrate solution is applied for from 1 to 2 hours. This solution is prepared as follows: To 10 cc. of a 20 per cent silver nitrate solution add about 8 drops of 40 per cent NaOH to precipitate the silver. Then add, drop by drop, concentrated ammonia until the precipitate is just redissolved. It is preferable that there should be rather a slight excess of silver hydroxide than an excess of ammonia. Fill up to 50 cc. with distilled water. There should be no noticeable odor of ammonia from this rather strong solution. After washing in distilled water for about 30 seconds transfer into the reducing fluid, composed (according to Bielschowsky's method) as follows:

Seignette salt, 10 per cent	75 cc.
Formol, conc.	5 cc.
Dextrose, 30 per cent	75 cc.
Potassium carbonate, 10 per cent.....	20 cc.

Reduction takes place within a few minutes and may be watched under the microscope. The preparation may be left in the solution for several hours. Then wash thoroughly and, after passing through the alcohol series and xylene, mount in Canadian balsam.

The nerve processes stain dark brown to deep black on light yellow background; the finest branches are visible. Fibrocytes stain a light tan, their nuclei dark brown and the nucleoli black. Incidentally the cultures may become overstained, the clot assuming so dark a brown as to prevent microscopic observation in ordinary light. To remedy this, I found it advisable to treat the preparations, after washing out the reducing fluid, with a strong solution of nitric acid. Start with a 10 per cent solution; if this does not lighten the stain within a few minutes, apply a 15 per cent solution, or in case the stain is very dark, a 20 per cent solution. The nitric acid 'differentiates' the preparation by taking out the stain exclusively from the clot, leaving the nerves quite black. It sometimes takes an hour or more to 'bleach' the clot. As soon as the clot has become sufficiently transparent (check under the microscope), destaining is stopped by transferring the preparation into a solution of ammonia, leaving it therein for several minutes. The treatment with strong nitric acid, though seeming very crude, gave results which were in no way less brilliant and clear than those obtained with properly stained preparations.

EXPERIMENTAL

I. Mechanical factors

1. *The orientation effect.* As mentioned above it was the purpose of this series of experiments to test whether or not mechanical structures or ultrastructures in the medium such as were found to be the guiding principle in the outgrowth of fibrocytes play a corresponding role in directing the outgrowth of nerve fibers.³ By mechanical structures we have

³ It has become a custom to call the processes growing out from a neuroblast in vitro 'nerve fibers.' Although they actually represent only the axonic part of a nerve fiber, the term 'nerve fiber' will be applied to them in this paper for convenience.

in mind fibrillar or ultrafibrillar aggregations of polar colloid particles (micellae) in a definite orientation. Potent factors bringing about such structures in an unorganized colloid substratum are oriented mechanical forces such as are exhibited, for instance, by stretching, pressing or stroking, or sending currents of fluid through, the medium. These factors, when acting for some time on a colloid composed of rod-shaped micellae in a constant direction, cause the micellae to orient and to associate with each other along the lines of force.

In the earlier experiments with fibrocytes (P. Weiss, '29), the orienting effect was produced by definitely distributed tensions, originating under certain conditions in thin membranes. In a liquid membrane stretching itself out between the sides of a horizontal triangular glass frame, the main tensions are directed at right angles toward the sides of the frame; a corresponding orientation is, then, assumed by fibrocytes growing out in such membranes. Since this method of creating tensions and corresponding structures of a definite pattern had proved so successful in the work on fibrocytes, it seemed advisable to start the nerve work by the same method. However, it very soon became clear that this method, if not altogether failing, was far less adequate for nerves than for fibrocytes. The nerve sprouts in brain cultures in the 'membrane in the frame' grew in practically all directions without showing any trace of the orientation effect. This was always the result if no other types of cells were contained in the culture than nerve cells. Spinal ganglia, on the other hand, gave, in several instances, slightly positive results; in these cultures, where fibrocytes and sheath cells were mixed with nerve cells, the nerve fibers sometimes showed a preferential course in those directions in which major tensions were supposed to exist.

The difference in reaction of pure and of mixed nerve cultures furnished a clue for explaining the failure of the method in the former case: It was noted that such tension as is necessary to form leading structures in the medium does not

arise unless there appears some additional contraction within the medium, and that such contraction is always associated with, and dependent on, some dehydrating activity of the culture exerted upon the medium. In my work dealing with fibrocytes, I have demonstrated this activity (the contraction of the clot has since been recognized by v. Möllendorff, '32, also) and I have shown that this activity is strictly connected with the presence of proliferating cells. Dehydration and contraction and the increase of internal tensions is not observed unless the cells of the culture multiply; merely being alive, without proliferation, will not produce any effect. Now, there is no proliferating activity whatsoever to be found in pure brain cultures during the first days, which are of concern in our experiments; the neuroblasts differentiate and send out processes, but do not divide. Proliferation being absent, the intensification of tensions to the point of effectiveness in creating structures in the membrane fails to take place, which explains the failure of pure nerve cultures to show any oriented growth in such membranes. It is not that the nerve fibers would not follow the mechanical structures established in the medium, but the supposed structures had simply not been established.

The positive results obtained with spinal ganglia, the non-nervous cells of which multiply, confirm this view. Besides helping to establish the necessary tensions, the use of spinal ganglia offers another advantage, in that the main pattern of the outgrowing sheath cells and fibrocytes, according to our previous experiences, is a sort of reproduction of the invisible structural pattern of the medium. Therefore, if we find the nerve processes oriented parallel to the nearby sheath cells and fibrocytes, we must conclude that the course of the nerve fiber has also been determined by the structural pattern of the medium. Since in a number of cases the nerve fibers were found to run parallel to, but independent of, spindle cells in the neighborhood, such a conclusion seemed well substantiated. Mechanical structures in the medium apparently served as guiding tracks for nerve processes as well as for spindle cells.

However, a more striking demonstration of this fact was desirable, and various attempts were made with different procedures. Finally, the following method proved satisfactory. A small drop of blood plasma, mixed with a small amount of embryonic extract, is put on a cover slip and spread well. Before coagulation sets in, one begins to stroke this layer in a constant longitudinal direction by means of a hair brush or needle. The stroking is continued until coagulation makes itself manifest by fine threads of plasma sticking to the stroking instrument. Then, by stretching the end which sticks to the brush from the other end which has become fixed on the cover slip, the clot is extended in a longitudinal direction and kept in this position until it adheres firmly to the cover slip, and finally the brush is removed. The tissue fragment is introduced into the medium either before or after the stretching. In order to secure better adhesion of the clot, mica cover slips are used.

Under the conditions of this experiment, nerve processes growing out from the explanted brain fragments do not follow the haphazard irregular courses which are observed in ordinary cover slip cultures of neuroblasts, but generally follow a course oriented in the direction in which the clot has been stroked and stretched. Figure 1 shows a typical case. One sees large strands of nerve processes which have grown out from the central fragment with an obvious tendency to orient in a given direction. Since the deeper layers of the clot, i.e., those closer to the supporting cover slip, are less affected by the stretching than the more superficial ones, the orientation effect does not appear in nerve processes which happen to take their course in those deeper layers (out of focus in the picture). The case shown in the illustration exemplifies the characteristic features of all the other cases treated by the same method. Figure 2 shows part of the fibers of a similar case under higher magnification; the fibers are growing from left to right, in the direction in which the medium has been stretched. In cultures of spinal ganglia the orientation effect is manifested by both the nerve processes and the sheath cells

and fibrocytes; all three run in the direction of the stroke of the medium, parallel to, though independent of, each other, save for occasional apposition (fig. 3). Their independence is proved beyond doubt by microscopical examination. Figure 4 shows a nerve fiber and a sheath cell traveling side by side in the direction of the 'stroke' of the medium without

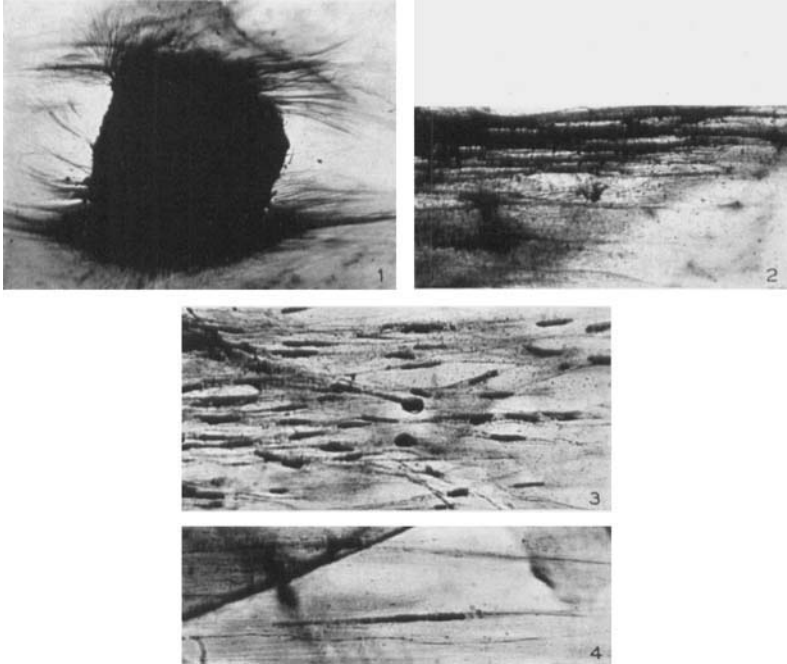


Fig. 1 Experiment 196. Fragment of medulla (sixth-day embryo), cultivated for 3 days in a stretched clot (direction of stroke horizontal in the picture). The fragment was inserted after coagulation of the clot. $\times 19$.

Fig. 2 Experiment 188. Cell-free part of a culture of spinal ganglion (twelfth-day embryo), cultivated for 2 days in a stretched clot. Direction of stroke from left to right. All nerve fibers oriented. $\times 170$.

Fig. 3 Experiment 222/1. Part of a culture of spinal ganglion (thirteenth-day embryo), grown for 3 days in a stretched clot. Direction of stroke from left to right. Spindle cells and nerve fibers oriented in the direction of stroke. Structure of ground-substance faintly visible. $\times 230$.

Fig. 4 Experiment 222/1. One sheath cell and two nerve fibers from the same culture as in figure 3. The oriented structure of the ground-substance is recognizable. $\times 465$.

touching each other. Although a nerve fiber which comes to touch a sheath cell will keep creeping along it, there has never any indication been found of the existence of any sort of specific attraction of the nerve processes by the sheath cells.

Another important fact is illustrated by figure 4. It will be noticed that the course of the nerve fiber is somewhat wavy, at least not as straight as the sheath cell nearby. This is a general experience, which holds for all nerve processes that grow out freely. Figure 2 furnishes further examples. This slightly wavy form of the nerve fibers is certainly not due to shrinkage of the medium during fixation, since it can be observed in living preparations as well. Fibrocytes, on the other hand, and especially sheath cells, if migrating in a medium with longitudinal structure such as the one we are dealing with, generally appear perfectly straight (compare Weiss, '29, fig. 18, p. 474). To understand the difference in the appearance between fibrocytes and nerve fibers, we must consider the difference in their mode of growth. The fibrocyte migrates out as a whole, its advancing tip dragging the cell body behind, under some tension which obviously suffices to straighten out the shape of the cell. The nerve cell, on the other hand, remains anchored in the central explanted fragment and pushes its processes into the medium; fresh substance is continuously interposed between the cell body and the growing tip, hence no stretch is present to straighten the fiber out. The situation changes at once if the nerve fiber becomes attached to some migrating sheath cell or fibrocyte. In this case, the migrating sheath cell, by its very movement away from the fixed body of the nerve cell, exerts some pull on the adherent nerve process, thus straining and straightening it out.

The same passive straightening of nerve fibers is observed, whenever there is some proteolytic liquefaction occurring in the medium after the nerve fibers have grown out. Such liquefaction is frequently found to take place on the second day after explantation, around the edge of the explanted brain fragment, especially if this latter contains parts of the

ependymal layer (P. Weiss, '34). If, then, the solid parts of the medium which surround the liquefied area retract from the explant, all nerve fibers that happen to lie across the gap become stretched and appear perfectly straight (compare Olivo, '27, '28). Another condition under which the nerve fibers appear straight or only slightly curved is found in their growth on the surface of the clot. There is always a certain amount of liquid covering the clot, and it is probably due to the surface tensions at work there that the network of those nerve fibers which grow on the surface is stretched out. If the clot is mounted on a glass cover slip instead of one made from mica, there is usually a certain small amount of fluid accumulated between clot and glass, and the nerve fibers which happen to penetrate to that depth show the same characteristic straightness of course as those growing on the outer surface of the clot. Moreover, nerve fibers which are grown in liquid media on the surface of a cover slip, as in Lewis' method, frequently become quite straight. That it is really surface tensions acting on phase boundaries such as are represented by the surfaces of the clot and of the cover slip, which result in the passive straightening of the nerve fiber, is further evidenced by the peculiar tendency to associate, anastomose, and group into bundles, as exhibited by the fibers lying in such surfaces, in contradistinction to the singular, independent and isolated course taken by those growing inside of a solid clot. Such an aggregation is generally a reliable indicator of surface tensions at work.

All these data prove that in order to straighten the nerve fiber some passive stretch, exerted either by elastic or by surface tensions, is required, whereas the active growth form which the nerve fiber achieves by its own tendencies, by its growth pressure so to speak, is somewhat bent and wavy, and comparable to a soft wire which has been driven into a substance of some resistance. We may also reverse this statement, and conclude that, wherever a nerve fiber is found in the characteristic wavy form, there is no tension acting upon it. Therefore, the wavy nerve fibers which in our experi-

ments have taken a definitely oriented course, can by no means be assumed to have done so under the direct influence of some sort of pull, but must have actively chosen such course. Huzella ('29), in dealing with fibrocytes, is inclined to consider tensional pull as an immediate factor in oriented growth. The real situation, however, is that the tensional forces act primarily upon the medium by provoking therein the formation of oriented structures, and secondarily these structures serve as guiding tracks for the protruding, pioneering tips of the cells or nerve processes. Accordingly, in our present experiments, the stroking and stretching of the blood plasma in one definite direction has caused the micellae to orient and aggregate in the given direction so as to establish by their parallel arrangement a definitely oriented structural pattern. Presently, the nerve cells begin to send out their processes. By virtue of certain properties which are not clearly analyzed as yet and are conventionally called 'thigmotactic,' the growing tip of the nerve fiber tends to orient and progress along the solid micellar aggregations of the medium. Such aggregations all being oriented in nearly the same direction, a course approximately parallel is imposed upon the numerous outgrowing nerve tips. Depending on local alterations of the underlying pattern, minor aberrations from that course occur, which, the central end of the fiber remaining fixed, explain the wavy appearance the whole fiber finally offers.

We have so far considered only the orientation of the nerve processes. A glance at figure 1 shows, however, and other cases confirm it, that the length or, in other words, the rate of growth of the nerve fiber is affected as well. Those fibers which take an almost straight course along the organized parallel structures of the medium are, after 2 or 3 days' growth, markedly longer than those which run through unorganized parts of the clot. This result could be explained by the greater resistance and the greater number of obstacles met with in the unorganized regions, as compared with the unhindered gliding along the organized tracks. However,

judging from what the earlier experiments with fibrocytes have revealed, this explanation may be too simple. In the experiments with fibrocytes it has been demonstrated that the better supply of liquids in the organized parts, and not merely the lack of resistance, is responsible for the acceleration of growth (P. Weiss, '29). Although no experiments are yet at hand to prove that a similar explanation holds true for the nerve fibers as well, this possibility should be kept in mind (compare below).

The facts revealed by this first set of experiments may be summarized as follows: By mechanically (stroking, stretching) evoking the formation of oriented ultrastructures (micellar aggregations) or structures (fibrillae) in the plasma clot, a corresponding orientation can be forced upon the nerve fibers growing therein. The course of nerve processes, therefore, appears to be strictly dependent upon, and directed by, the mechanical organization of their environment.

2. *The formation of connecting tracts between two centers.* The following phenomenon has been described in the earlier work on fibrocytes (Weiss, '29, p. 540). If two cultures somewhat distant from each other are put in one of those extremely thin 'plasma membranes in the frame,' an extensive growth of cells appears along the connecting line of the two cultures, as if each fragment were attracting the cells of the other. However, what at first glance appears to be the action at a distance of some undefined attraction can easily be explained on an essentially mechanical basis, without involving any hypothetic 'histiotropic' forces or the like. It can be shown that merely the presence of two active growth centers necessarily results in the formation in the medium of a structural pattern of such arrangement as to offer to the cells a straight connection between the two centers as a preferential pathway.

Since, in the course of the present experiments with nerves, rather extensive use has been made of this method, a more thorough description of the experimental procedure is advisable than was given in my earlier paper. It has been men-

tioned above that a proliferating culture has some dehydrating effect upon the surrounding medium. Under that influence the clot tends to solidify while some of its capillary fluid is set free; part of the fluid released is utilized by the growing cells and the rest can be found accumulated on the surface of the clot in the neighborhood of the culture. The result is that, in a localized area around the culture, the plasma contracts and shrinks. Conspicuous traces of such local shrinkage are quite commonly found in standard cover slip cultures of fibrocytes as a fairly large depression that forms right above the culture in the clot after a few days' growth. The local contraction brings about tensions in the rest of the clot which center toward the contracting area and which are the more intense the smaller the volume of the clot; in extremely thin clots, such as are represented by the 'membranes in the frame' in our experiments, they must amount to very high intensities. Now let us consider a case where there are two such growth centers in a membrane, both proliferating, both dehydrating their surrounding medium, and thereby establishing two centers of contraction; what must be the system of tensions arising in the membrane under these conditions?

The situation can be easily reproduced by means of a model: a piece of coarse gauze or some other netted fabric is spread over a disk of cardboard and fastened along its circumference. This is to represent the plasma membrane in the frame. By pulling simultaneously through two holes that have been made in the cardboard as much of the fabric as its elasticity permits, we imitate the two local contractions. Consequently we find the meshes of the fabric distorted in a definite way, the resulting pattern giving a true picture of the pattern of the tension trajectories that are created by two local contractions in a membrane. Bearing in mind further, that the tension trajectories give rise to a corresponding pattern of ultrastructures, we recognize the meshwork in its distortion to be a true image of the structural organization arising in a membrane in which two cultures, somewhat distant from each other, are proliferating. A glance at the

picture thus obtained (fig. 5) at once reveals strikingly that the most pronounced structure is a dense pattern of almost parallel lines, connecting the two centers; we may call this characteristic pattern of micellar orientation along the connecting line between two proliferating cultures the 'bridge.' The continued suction on the part of the two growing cultures causing a steady shifting of liquid along the once established lines, tends to, and succeeds in, elaborating and accentuating still further those bridge-like communications. The cells from both ends enter the bridge and move along it, as they would do along every structure of that sort. In this way, they are finally bound to reach the opposite center, which is likely to give the false impression that something emanating from the opposite center had attracted them.

According to previous experiences it might be expected that provided any 'bridge' was established in the medium between two nerve cultures, nerve fibers, like fibrocytes, would follow it. The numerous experiments that were made in this series gave quite consistent results. The method was as described in the fibrocyte work (P. Weiss, '29, p. 446). Small glass frames with sides of about 1 cm. in length (circular, square or triangular) were dipped into blood plasma, then withdrawn horizontally. Thus, a thin membrane of blood plasma remains stretched across the frame like a window. By adding a little embryonic extract the membrane is made to coagulate, after two, or in some cases three, tissue fragments have been introduced. The frame is then placed in a small glass bulb, containing some water to keep the air moist. Again, both brain tissue and spinal ganglia were used, cell proliferation being absent in the pure brain cultures but very extensive in the spinal ganglion cultures on account of their content in sheath cells and fibrocytes. The cultures, which were less than 1 mm. in diameter were put in the membrane at a distance of from 2 to 6 mm. from each other.

The results after 3 days' cultivation in the incubator were uniformly as follows: A narrow strand of tissue connecting the two cultures along a straight line developed in every case

where both cultures were spinal ganglia, never between pure brain cultures. Figure 6 shows two spinal ganglia and the connecting tract of cells and fibers. Under higher magnification (fig. 7) we can distinguish spindle cells and nerve fibers, both utilizing the same system of parallel pathways. In contemplating figure 6 (low magnification) one is struck not only by the definite, sharp orientation of the cells and fibers along the organized 'bridge,' but also by the fact that they have grown much farther in that particular direction than in any other, which holds true for all other cases as well. This falls in line with the observations made in the fibrocyte experiments which have shown that growth is always enhanced in the 'organized' regions of the clot, i.e., in those regions in which definitely oriented structures have been provoked (Weiss, '29, p. 498). Evidence is at hand and has been described, indicating that the promotion of growth along the 'organized' zones is chiefly due to a better and easier supply to the cells of the nutritional and growth-promoting liquids contained in the clot, as a result of mechanical peculiarities of the 'organized' areas themselves; along the oriented struc-

Fig. 5 Model, showing the orientation of tensions arising in a system in which two centers of contraction exist.

Fig. 6 Experiment 220. Two spinal ganglia (thirteenth-day embryo), cultivated for 3 days in an extremely thin plasma membrane. A straight tract of spindle cells and nerve fibers has formed between the two cultures. $\times 31$.

Fig. 7 Experiment 220. Part of figure 6 under higher magnification, showing cells and nerve fibers traveling in parallel lines over the 'bridge' between two centers. Growth direction from left to right. $\times 140$.

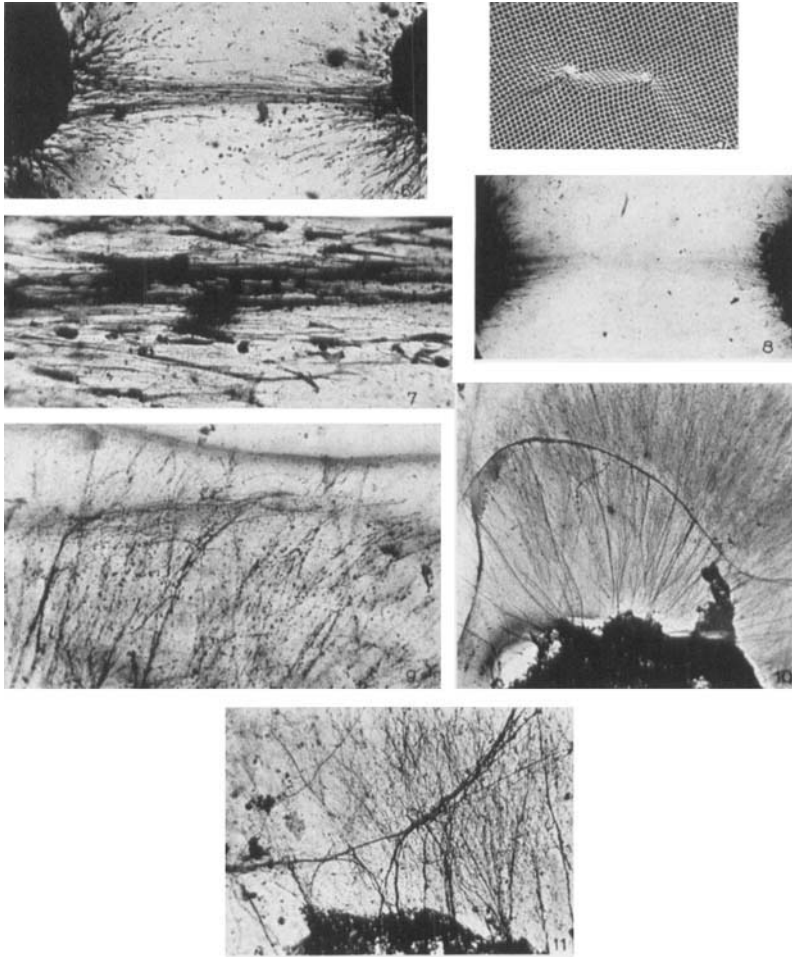
Fig. 8 Experiment 206. Two spinal ganglia (tenth-day embryo), cultivated under the same conditions as experiment 220 (fig. 6), but for 2 days only. The condensation of the medium along the connecting line between the two cultures, the 'bridge', can be recognized by its darker stain. $\times 22$.

Fig. 9 Experiment 176. Fragment of medulla (sixth-day embryo), grown for 2 days in a streak of diluted plasma which was surrounded by concentrated plasma. A nerve plexus has developed along the boundaries between the two sorts of plasma. $\times 113$.

Fig. 10 Experiment 179. Formation of a peripheral plexus by nerve fibers (medulla, sixth-day embryo) grown for 2 days next to the surface of the clot, the culture exhibiting a slight degree of liquefaction. $\times 43$.

Fig. 11 Experiment 179. Part of the plexus shown in figure 10, under higher magnification. $\times 195$.

tures retention of liquids by the clot is weakened and their shifting toward the culture is facilitated (Weiss, '29, p. 528). If, in nerve fibers, as in fibrocytes, the growth rate depends



Figures 5 to 11

upon the facility of securing an adequate local supply of liquids, it may be unnecessary to assume that any specific substances should emanate from the two cultures which were

responsible for the accelerated growth. Thus, both the directive and the growth-promoting effects which lead to the formation of straight connecting tracts of cells and fibers between two growth centers can be satisfactorily explained by the mechanical, i.e., structural, organization of the medium, by the formation of the 'bridge.' Still it might be questioned whether this explanation is of more than hypothetical value: Is it just a conception the simplicity of which renders it a useful working hypothesis, or are there other facts to prove its reality? There are.

In considering any one of the membranes containing two spinal ganglia on the second or third day of cultivation, one invariably notices that the strip of plasma that lies between the two cultures has become quite different from the rest of the membrane, in that it appears much denser, less transparent, and in the stained preparation assumes a much darker stain; this can be recognized, though only faintly, even in the photograph, figure 6. A continuous streak can be seen reaching from one culture to the other, long before the cells and fibers growing out from either end have come to meet in the middle. Figure 8 is an attempt to show the appearance of that streak at an early stage, when growth had not yet proceeded as far as in figure 6. The streak of denser plasma, leading straight from one culture to the other, and in many cases even revealing longitudinal parallel structures, is nothing else but the 'bridge' which we have been commenting on made manifest.

Furthermore, beyond the positive evidence furnished by the visibility of the 'bridge,' negative evidence can be obtained from almost every case, disproving the assumption that the cells and fibers were directly attracted by 'chemotactic' influences emanating from the cultures. By diffusing chemotactic substances as large a part of the edge of each culture would necessarily be reached and affected as is facing the opposite culture; that would be almost half of the circumference, and all the cells and fibers emerging from that half ought then to show orientation toward the opposite culture. This is ex-

actly what they do not show. It is quite clear from figure 6, as from many other cases, that only within a limited and narrow fraction of the whole length of the borders facing each other, the cells and fibers converge upon the 'bridge,' while those originating from the rest of the edge do not show any orientation effect at all, although their position is such as to directly expose them to whatever substances might diffuse from the opposite culture. On the other hand, from the standpoint of the explanation we have offered, it is easy to understand why the orientation effect shows only in a comparatively small region. The development of the 'bridge,' as we saw, results from local dehydration, associated with proliferation. Therefore, the points for the 'bridge' to start from are those where proliferation is most active. In fact, figure 6 readily shows that the 'bridge' originates from those parts of the fragment that are most crowded with cells. This same situation is encountered in every case where proliferation does not go on at an equal rate all along the edge of the fragment but is confined to certain localized clusters of cells. It is evident that this striking feature speaks in favor of the 'structural' against the 'chemotactic' explanation.

The final and perhaps the most convincing evidence is, however, furnished by the results with pure brain tissue: In no case where there have been two fragments of brain instead of two spinal ganglia opposed to each other in a membrane, has any trace of mutual influence, with reference to orientation and growth rate of the outgrowing nerve fibers, been observed. The fibers emerging from the brain fragments spread out quite irregularly without showing the slightest tendency to deviate from their haphazard courses; they do not orient toward the other culture, not even when they incidentally reach its immediate neighborhood. The resulting irregular pattern of the nerve fibers is not different from what it would be had both fragments been cultivated each in a separate membrane. Now, the chief difference between a culture of spinal ganglia and one of pure brain is the presence of proliferating cells in the former and the complete absence

of such in the latter. Accordingly the dehydrating activity associated with proliferation, as exhibited by spinal ganglia, is lacking in brain cultures. Since, as stated above, leading structures or 'bridges' connecting two cultures arise only if both cultures keep dehydrating their surrounding medium, no 'bridge' can be expected to form between two brain fragments. In consequence, the nerve fibers under these conditions fail to orient toward the opposite culture, which proves beyond doubt that, for their orientation, they depend on the presence of underlying oriented structures, and not upon some 'chemotactic' substance which would attract them from a distance; for it is reasonable to assume that such substances would be given off by brain cultures as well as by spinal ganglia.

Summarizing the results of section 2, we may say that the formation of oriented structures guiding the nerve fibers can be provoked by a certain activity of the cultures themselves, provided the latter contain proliferating cells. Development of nerve fiber tracts forming straight connections between two growing cultures is explained on this basis.

3. *The formation of plexuses.* If nerve fibers on their way through the clot happen to encounter a region with a differently oriented structure, we may expect them to deviate from their original courses to follow the new orientation. They will continue to run along the new path until some other influence deflects them again. Whether a fiber in entering the cross path will turn right or left will depend on its angle of approach, both ways being equally accessible. Fibers that previously have been running separately and independently will be brought together on cross paths and will interlace with each other to form a new arrangement. In other words, what we would call, in the organism, a nerve plexus is being formed.

The situation, as here described, was realized in the following experiment: Various experiences show that in a plasma clot which consists of two distinct parts of differing mechanical constitution, structures arise along and parallel to the boundary in which the two sorts of plasma fuse. By mechani-

cal constitution we mean the greater or lesser firmness of the clot which can be controlled by varying the dilution of the plasma. It must be the vigorous and steadily increasing tensions in the surfaces of a clot which cause the micellae in the external layers to orient in sheets parallel to the surface. Apparently the formation of a sort of ultramicroscopic lamellar 'barrier' along the internal surface separating two different, though intimately fused clots must be referred to the same forces. If, then, nerve fibers are allowed to grow out from one part of the clot toward the other, those which happen to strike the 'barrier' are deflected from their way and become interlaced so as to give rise to a plexus-like formation.

Numerous experiments of this kind were made. First a narrow streak of blood plasma of a given concentration is put on a cover slip and a brain fragment or spinal ganglion placed therein, not more than 2 mm. from the edge. Coagulation is effected by touching the plasma with a needle that had previously been dipped in embryonic extract. After about half an hour, when coagulation is perfect, the first clot is surrounded by a second clot of different concentration. Various grades of concentration are obtained by diluting the concentrated plasma with Tyrode solution. In one set of experiments, the central clot was made of concentrated plasma and the surrounding clot of diluted plasma, while in another set the reverse arrangement was effected.

Nerve fibers, growing out in all directions from the culture contained in the central clot, reach the edge of their clot on the second or third day of growth. There they encounter the structures of the 'barrier' which divert many of them from their more or less radial orientation into a tangential course. Some of the fibers, then, keep running along the 'barrier,' others, after a time, penetrate into the medium of different consistency beyond, some are entirely arrested in the 'barrier,' and still others run right across the barrier over to the other side. To the last group belong especially those fibers that arrive at the 'barrier' at right angles. Generally one gets the impression that the smaller the angle is, under which a fiber approaches the barrier, the more likely the fiber is to

be deflected in a tangential direction. Figure 9 shows a typical case. The 'barrier' which can be recognized by the darker stain runs horizontally through the upper part of the picture, separating the more dilute plasma (below) from the more concentrated plasma (above). The nerve fibers (brain culture) come from below, the general direction of growth being upward. The formation of the plexus along the 'barrier' is almost schematically clear. There were several cases of this sort obtained, but not all of them were suitable for photographic reproduction. In cases like the one shown in the photograph, where the nerve fibers run from the softer medium against the denser one, it might be argued that it is merely the higher resistance met with in the more compact medium that caused the fibers either to be deflected from their course or to be stopped altogether. However, the result is exactly the same in those even more numerous cases where the fibers run from the denser against the softer medium, i.e., from higher to lower resistance; in those cases also the fibers are again deflected to form a plexus or are completely stopped in their course, just as in the former case. This proves that it is the structural organization of the 'barrier' itself, and not the resistance offered by the medium beyond it, that causes the plexus formation.

Again, it is possible, instead of artificially creating the conditions for plexus formation, to let the culture itself establish such conditions. It has been mentioned above that brain fragments frequently tend to liquefy their surrounding medium. I have found that this proteolytic activity is confined to the ependymal layers (P. Weiss, '34). In cultures consisting exclusively of ependymal tissue, the liquefaction is so vigorous and so extensive that complete destruction of the clot results around the fragment, leaving the culture floating in a liquid area. If brain tissue is used that has been carefully stripped of all ependymal fragments, no liquefaction occurs. Therefore, in a brain fragment excised without discrimination the intensity of liquefaction will depend on the amount of ependymal tissue included in the fragment. This amount being, in some cases, small enough to prevent complete destruction of

the structure of the clot, but sufficient to produce some slight change, a situation arises similar to that brought about by the use of different dilutions of plasma in the experiments reported above. By the slight liquefying activity spreading from the fragment, an effect equivalent to dilution is obtained; in restricted, especially superficial, parts of the plasma around the fragment, the content in capillary liquid gradually increases, without going as far as to entirely abolish the existing solid structures. The area of dilution is set off from the unaltered remainder of the clot by a remarkably sharp outline, thus giving an opportunity for the development of a 'barrier' structure along the border. Accordingly, those nerve fibers which grow out subsequently are forced to form a plexus as soon as they reach the border. Such is the case in figure 10. The brain fragment is surrounded by the 'diluted' area (perceptible by its somewhat lighter stain), the edge of which is occupied by a conspicuous black plexus of nerve fibers. Figure 11 shows part of the plexus under higher magnification, clearly illustrating how nerve fibers (coming from below) fuse with, and again emerge from the plexus.

Summarizing the results of section 3, we may state that plexus formation is the result of the presence in the medium of an area with oriented structures running across the original pathways of the nerve fibers.

II. Chemical agents

Although it is quite obvious from the experiments described in the first part of this paper that the immediate agent directing the course of the nerve fiber on its way through the clot is mechanical—the structural pattern of the clot—it must be admitted that, as an indirect factor, chemical activities of a certain kind play an important part. They contribute to the establishment of the leading mechanical pattern, sometimes even determining its essential configuration. In our experiments we have encountered two sorts of chemical activities which proved to be potent factors in the formation of structural patterns in the clot: one is dehydration, essential, e.g., for the formation of 'bridges' and fiber tracts between growing

centers; the other is proteolytic liquefaction, responsible under certain circumstances for the formation of 'barriers' and nerve plexuses. There can be little doubt that many other sorts of similar activities influence, in one way or another, the structural organization of colloidal substrata. But still, as far as the nerve fiber is concerned, their influence is an indirect one, they work on the medium and not on the fiber.

Such an indirect influence is not, however, what one has in mind when speaking of 'chemotactic' effects. The term rather implies that direct influences emanating from a chemical source reach the nerve fiber and cause it to orient and to keep definitely oriented toward the source (compare Tello, '23). As we have previously stated, in brain cultures no trace of any direct chemical 'attraction' can be detected under conditions which would necessarily have revealed it, were it existent. One might object, however, to plasma membranes such as were used in these experiments, on the basis that the mechanical structures might have been too strong and the chemotactic activity of the brain fragments too weak to allow the latter to manifest itself. To check on this possibility a set of experiments was carried out in which a clot of diluted plasma on a cover slip was substituted for the membrane, and the brain fragment was replaced by concentrated brain extract, presumably a stronger 'chemotactic' source. The brain extract was obtained by crushing the whole brain of a 14-day-old embryo in a few drops of Tyrode solution and centrifuging several times for 5 to 10 minutes. A brain fragment was then 'confronted' with the open end of a capillary tube filled with brain extract, culture and tube being included in a common clot on a cover slip (comparable to Centanni's, '14 'colture affrontate').⁴ In experiments of this sort, the tube should be

⁴It is impossible to tell from Centanni's representation of his experiments (cultivation of two fragments of different tissues side by side) to what extent structural effects may have simulated primary chemical effects; but, to judge from his method, it is clear that such mechanical effects were in play. Moreover, it must be pointed out that when Centanni refers to 'nerve growth' he means growth of the spindle cells contained in the nerve and not of nerve fibers, since he has used fragments of peripheral nerve only, from which there is no outgrowth of nerve fibers.

completely submerged in the clot, otherwise a 'chemotactic' effect can be simulated by what is really a tension effect. This point will be further considered below.

The results of the experiments were uniform: In none of the numerous cases did the nerve fibers growing out from the brain fragment converge, or even show an inclination to orient, toward the end of the tube as the center of the diffusion of the brain extract, nor was their course and pattern—the irregular mycelium-like pattern of ordinary cover-slip cultures—affected in any other perceptible way. The experiments were repeated with the tubes filled with muscle extract instead of brain extract. The results were again strikingly negative. Then, in some experiments, a brain culture was confronted with a fragment of peripheral nerve of an older embryo. No fibers were 'attracted' by this source. Experiments in which brain cultures were confronted with fibrocyte cultures, embryonic muscle, skin fragments, myotomes, were likewise negative.

Now, none of the agents above mentioned, neither brain extract, muscle extract, freshly excised peripheral nerve, nor cultures of embryonic organs, have ever been demonstrated in a convincing manner or been maintained to be sources of 'chemotactic' stimuli. What has always been claimed, however, to be a potent source is the peripheral nerve in the state of degeneration which follows severing its connection with the centers. Cajal ('08) and Forssman (1898, '00) have repeatedly asserted that there are influences spreading from the cut end of a degenerating peripheral nerve which serve to attract new regenerating fibers from a distance; though the experimental facts to prove the existence of such 'neurotropism,' as it was called, are very suggestive, its interpretation as a sort of 'chemotaxis' is merely hypothetical and open to question. It was with this interpretation in mind that the following experiment was devised: In an adult chicken (1 year of age) one of the wing nerves was severed distally from the shoulder plexus. Eight days later, when the degeneration of the peripheral stump could be assumed to be well under

way, small parts were excised from it. With each one of the nerve fragments a brain fragment (or spinal ganglion) was confronted in a cover-slip culture, the cross section of the nerve facing the brain fragment. Yet again, the nerve fibers growing out from the brain fragment did not appear in any way oriented or otherwise affected by the degenerating nerve fragment. It is certainly disappointing to find the 'neurotropic' agent that has been supposed to be the most efficient—the degenerating peripheral nerve—failing entirely in the first direct test to which it has been subjected. There can be no doubt that if there were chemical substances of neurotropic capacity diffusing from a degenerating nerve, their effect would have become noticeable under the conditions of our experiment; therefore, these experiments seem to disprove the assumption of 'neurotropism' being a sort of 'chemotaxis.' The same conclusion was reached by Dustin ('10) for regenerating nerves.

An incidental observation in one of our experiments is very illuminating in that respect: In a particular case the amount of plasma employed was too small to completely include and cover the explanted pieces, a spinal ganglion and a fragment of degenerated nerve, which, therefore, projected out of the clot, considerably deforming its surface. When subsequently the outgrowth of cells and nerve fibers from the spinal ganglion took place, those growing near the surface of the clot proceeded in a straight course toward the nerve fragment, quite as if they were attracted by it. Focusing to a deeper level, however, disclosed that the mass of fibers which were growing inside the clot failed to show any definite orientation with regard to the nerve fragment. Were the orientation effect due to a direct chemical stimulation, it could not well have remained restricted to the surface, but would have operated in the whole diffusion field, i.e., in the interior as well as on the surface. A mechanical explanation, however, is readily offered by the peculiarities of the mechanical conditions which are different on the surface and in the interior. The surface, in some way, corresponds to the membrane in the experiments

of section I. 2; it is clear that dehydration and contraction resulting from mere evaporation take place at a much higher degree on the surface than in the interior. Now, there are two points which remain unalterably fixed during the gradual contraction of the surface, i.e., the points where the surface has been raised by the two fragments. It is easy to realize that at the surface, then, a bridge-like structure connecting the two outstanding points must arise, which accounts for the orientation of the superficial fibers, whereas the more homogeneous conditions inside the clot explain the absence there of such orientation. Apparent 'neurotropism,' in this case, therefore, is nothing but the outcome of certain structural effects in a particular situation. Many more situations of a similar kind can be visualized in all of which the mechanical orientation of the nerve fibers could easily be mistaken for a chemotactic effect, the more so, as very frequently chemical activity of some sort is an essential component of such situations.

Summarizing, then, the results of this section, we can state that the facts are strictly contradictory to the assumption of a direct 'chemotactic' orientability of nerve fibers. To be sure, centers of chemical activity may be, as has been described above in chapter I. 2, of determinative significance in the orientation of nerve fibers; however, in an indirect way only, if and provided their activity is such as to affect the structural pattern of the substratum.

III. Electrical factors

Electric influences have frequently been assumed to be orienting factors in nerve growth (Kappers, '17; Child, '21), but the only serious attempt to verify that hypothesis by experimental factors is the work of Ingvar ('20). He exposed nerve cultures in vitro to electric currents and claims to have obtained a definite orientation of the nerve fibers with respect to the electrodes. Unfortunately, only a short preliminary account of these experiments has been published from which only very incomplete information as to methods and results

can be gained. Nevertheless, when I started out to repeat these experiments, I was fully convinced that Ingvar's main result, such as it was, could be taken as a matter of fact, although the interpretation I was inclined to give the results was somewhat different. Knowing that oriented vectorial forces can be of orienting influence on the micellae of blood plasma, it seemed that the influence of the electric current on the orientation of nerve fibers might again be an indirect one, operating by the establishment of leading pathways oriented along the electric lines of force. This interpretation being as compatible with Ingvar's statements as the idea of a direct action of the current on the nerve fiber, further experiments were needed to decide the matter.

It was disappointing to find that neither alternative could be substantiated, as I have been unable to duplicate Ingvar's basic results in spite of exhaustive endeavors. For 1½ years ceaseless attempts were made to influence nerve fibers growing in vitro by electric currents; however, not the remotest indication of any success was ever detected. Nearly twenty different devices for applying the current have been used and a wide range of current intensities from the lowest applicable up to far beyond the lethal dose have been tested.

It is impossible here to refer in detail to all the methods that have been tried; only the main types will be mentioned: A sketch of the electric hook-up is given in figure 12. Currents of from 10^{-9} to 10^{-3} ampères could be obtained by this arrangement. A galvanometer was inserted in the circuit which, after interruption of the primary circuit, would indicate polarization currents originating from the culture. Two sorts of electrodes were in use: Silver wires coated with silver chloride, and mercury-calomel electrodes. The former were polarizable to a considerable degree. The latter (fig. 13) proved practically non-polarizable; from the calomel electrode the current was conducted to the culture by means of capillary tubes filled with Ringer-agar. In one set of experiments, the current was sent through a conductor (pencil lead) crossing (as in Ingvar's experiments) or perforating the clot (fig. 14). The galvanometer permitted a continuous check of

the flow of the current. The current was sent through the culture either permanently or temporarily; in most experiments it was applied after the medium had already coagulated (figs. 15, 16, 17). One series, however, was made in which the current was applied while coagulation was going on. Ex-

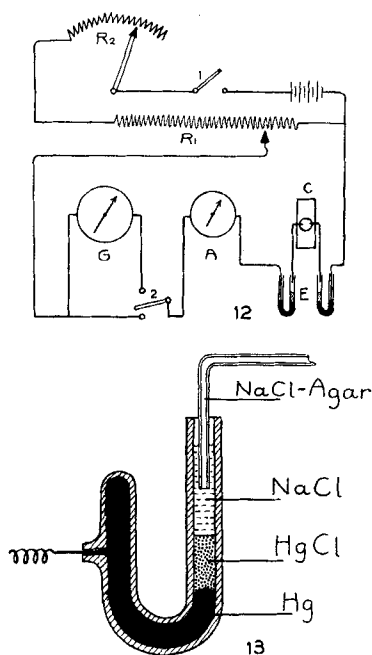


Fig. 12 Arrangement for the application of weak currents to cultures. $R_1 = 4300 \, \Omega$, $R_2 = 10,000 \, \Omega$. G, galvanometer; A, microammeter; C, culture; E, calomel-electrodes. By closing the switches 1 and 2 the current is sent through the culture, the intensity being controlled by means of the resistances R_1 and R_2 . By opening switch 1, while leaving switch 2 closed, any polarization current that might be produced in the culture can be disclosed on the galvanometer.

Fig. 13 Calomel electrode, as used in the experiments.

amples of the set up with explanations are given in the illustrations (figs. 14 to 17).

Finally, a few experiments were done in which the culture could be kept under continuous microscopic examination while exposed to the electric current. A cover-slip culture with the electrodes in place was mounted on a warm stage under the microscope. No current was applied the first day. On the

second day, the microscope was focused on a group of nerve fibers that had grown out at various angles and a drawing was made of those fibers. Then the current was sent through the culture and drawings were made at intervals. If no definite change was observed in the orientation of the growing

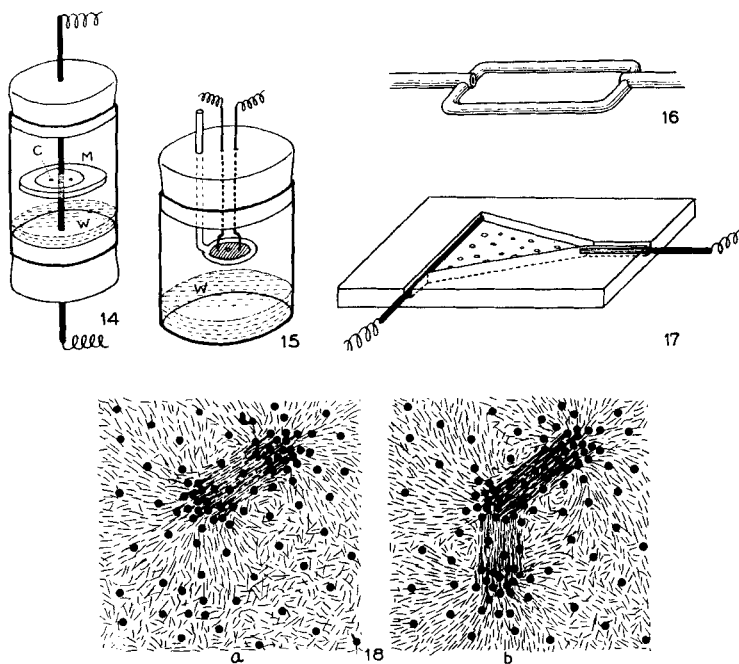


Fig. 14 Conductor (graphite) perforating a mica plate (M) carrying the culture (C).

Fig. 15 Culture kept in a thin plasma membrane between two silver-silver-chloride electrodes.

Fig. 16 Frame for holding thin plasma membranes, with side tubes leading to calomel-electrodes.

Fig. 17 Triangular glass trough for the application of currents of graded densities. One electrode runs along the base, the other enters at the apex of the triangle; the density of the stream lines is greatest at the apex and lowest at the base. The trough was filled with blood plasma and numerous cultures were scattered in it. A wide range of current intensities could, thus, be tested on a great number of cultures under otherwise identical conditions.

Fig. 18 Schematic picture showing the formation of connecting fiber tracts, a) between two centers of simultaneous development; b) between a younger (below) and a somewhat older center, b representing the same area as a at a later stage. The dots mark cells, the dashes indicate the supposed structural arrangement of the ground-substance.

tips of the fibers since the onset of the current the intensity of the current was diminished or increased. If there were any critical intensity on which the presumable orientation effect depended, it should have been revealed by this method of direct observation; the threshold values given by Ingvar were included in the range tested. However, again no positive result was obtained, the growing tips showing no tendency to orient toward either of the electrodes.

After all these strictly negative results it is hard to account for Ingvar's positive statements. Although there can be little doubt that he must have observed definite orientation effects of some sort, it is very doubtful that these were true electric effects. Incidentally Ingvar's statements, brief as they are, nevertheless contain a remark that makes the non-electric nature of his effects almost obvious: He states that when a conductor was laid across the plasma clot and an electric current sent through it, nerve fibers grew out at right angles toward the conductor. If they really oriented in this manner—and there is certainly no reason for doubting the correctness of Ingvar's observations—then this is the best proof that their orientation was not due to electric effects, for there are no vectors in the electro-magnetic field that would act at right angles to a conductor. From electrodynamic effects, orientation could result only in cylindrical surfaces around the conductor as the axis. A radial orientation, as observed in the nerve fibers, cannot be, therefore, an electric effect. On the other hand, it could well be a mechanical effect; if, for instance, the conductor projects out of the clot, tensions at right angles to the conductor arise in the surface of the clot, which would explain the result.⁵

⁵Ingvar considers this experiment as a test of Bok's hypothesis of 'fibrillogenous stimulation.' Bok ('15) showed that neuroblasts in the vicinity of a supposedly functioning nerve tract send their axones at right angles toward that tract. Without entering into a discussion of the matter, we simply wish to point out that Ingvar's assumption that he has reproduced in vitro the situation as visualized by Bok is quite erroneous; a functioning nerve fiber can never be reproduced by a conductor through which a current is sent; the propagation of a disturbance of the electric potential difference existing across the wall of the nerve fiber as occurring in nerve function and an electric current are two very different things indeed.

It remains to consider Ingvar's cases in which the nerve fibers are described as oriented toward the electrodes. One possibility, though rather a vague one, is the following: Occasionally I had the impression as if, opposite an electrode, the neuroblasts of the brain fragment were mobilized in such way as to abandon the close connection with the explanted piece and to scatter in the medium; fibers adhering to these migrating cells become stretched in the direction of the migration. Thus, nerve fibers would be passively oriented toward the electrodes. My observations in this matter are much too sporadic to allow any definite statement. Since the point is of interest in connection with Kapper's idea of the oriented migration of neuroblasts (neurobiotaxis, Kappers, '17) being due to electric influences, a more thorough investigation would be worth while. Of course, the migration of the neuroblast as a whole and the outgrowth of the nerve processes are quite different and independent phenomena and it is not impossible that the former might react to electrical influences to which the latter would not react. In any case, migration of the neuroblasts occurring opposite the electrodes is one way in which an orientation effect on the nerve processes could be simulated. Yet it is not very likely that this should have escaped Ingvar's attention. More suspicious is the sort of electrodes he used. From inquiries in this laboratory, where his work was performed, I gathered that he was using boot electrodes with wicks as leads. This, in my opinion, could well be the source of error. The slightest suction exerted by these wicks on the medium—and the balance of the fluids will never be so perfect as to prevent suction—must result in a flow of liquids oriented toward the points of suction, and the formation of a correspondingly oriented ultrastructure in the clot, a sort of 'bridge' (see p. 409) between the two sucking wicks, would ensue. This, too, is a mere suggestion. It is, of course, impossible to hold any definite opinion without knowing exactly the conditions of Ingvar's experiments. Still, it is important to point out that non-electric interpretations of his observations are feasible.

Notwithstanding the inconclusiveness of Ingvar's results and the negative outcome of my own experiments, the question remains open whether it would not be possible to force, by a more adequate application of electricity, perhaps by electrostatic fields, the micellae of a colloid medium into a definite orientation so as to form leading pathways.

DISCUSSION

The experiments reported in this paper certainly prove that oriented ultrastructures ('ultrastructural organization') in the medium constitute the most potent mechanism in nerve orientation, if not the only one. In fact, it seems doubtful whether other factors that one might regard as capable of orienting nerve fibers could work against those mechanical structures. As far as the tissue culture experiments are concerned, however, no factor of such sort has been found. For the conditions of tissue culture, therefore, the problem of orientation of the nerve fiber reduces itself to the problem of micellar orientation in colloidal substrata which is a purely physico-chemical problem.

The question now arises whether such a solution of the problem of nerve orientation would be feasible for the conditions in the organism also. In comparing the conditions of tissue culture with the conditions in the organism we eventually come to realize that, however different these conditions may appear to be, they do in their essential features correspond, in so far as the problem under discussion is concerned. To prove this, we recapitulate the main features met with in tissue culture experiments: The medium is a colloid taken from a normal organism; its elements (fibrin micellae) are polar rods. Oriented agents, acting on such colloid, force the rod-like elements into a definite orientation parallel to each other and to the directive agent. Mechanical tensions and persisting currents of liquids are such agents. Possibly any vectorial force may under adequate conditions be of orienting influence. The nerve fiber, then, owing to its tendency of progressing along microscopic surfaces or ultramicroscopic

interfaces, is bound to reproduce by its course the pre-established oriented pattern of the medium through which it happens to travel. Such are the conditions found in tissue culture. Now, as to the organism, the conditions are essentially comparable as I have recently pointed out in a general survey of the matter (P. Weiss, '33). There are colloidal substances present almost everywhere in the organism which are of the same order and show the same characteristics as the blood plasma in our tissue cultures. Their elements, high molecular protein compounds, are polar and rod-shaped. Being subjected to vectorial forces, mechanical or otherwise, they will arrange into oriented patterns which, if aggregation is allowed to continue, may eventually become visible in the form of microscopic fibrillae (v. Ebner, '06). These gelatinous substances, capable of assuming structures oriented along the lines of force, are present in the embryo as the so-called 'ground substances,' filling the spaces between the germ layers and in later stages between the various organs. This proves that a material, suitable to ultrastructural organization and comparable to the medium of tissue culture, is present in embryogenesis. The further fact that nerves tend to orient along such structures of the ground substance is clearly illustrated by the observations of Held ('09) and others. Held saw processes of neuroblasts, upon leaving the neural tube, follow fine fibrillar threads, bridging the space between neural tube, myotome and ectoderm. The threads were considered to be part of a continuous network known as Szily's fiber net. If we disregard their interpretation as plasmodesms, which has not been definitely proved (compare Snessarew, '32), we may consider them as traces of the pattern which we assume to exist in the ground substance, somewhat exaggerated by the process of fixation.

Every oriented ultrastructure of the living organism becomes coarser and more pronounced under the treatment of preservation, so that it may eventually attain microscopic visibility. Such is the case in Held's preparations, although this does not mean that microscopic filaments had not already

existed in Szily's fiber net in vivo. However, in general, the ultrastructure of the ground substance will not manifest itself under the microscope, so that in most cases the attempt to discover a leading substratum of microscopic visibility must fail.

Keeping this in mind will invalidate one of the main objections to the dominant role we attribute to underlying structures in nerve orientation, namely, the objection that microscopic structures to play this role are most commonly not noticeable. Tello ('23), for instance, has objected to His' (1887) notion that the agent which causes the maxillary nerve to split into a mandibular and a lingual branch, was the mechanical resistance of Meckel's cartilage. Tello states that at a time when the nerve has already effected its branching no histological differentiation is yet found in Meckel's cartilage. From our standpoint, such histological differentiation is not at all required to affect the course of the nerve since the ultrastructural arrangement of the pre-cartilaginous blastema, long before differentiation becomes manifest, is sufficient to deflect the nerve fibers.

This situation is further illustrated by our experiments on plexus formation, as described above. We saw that a nerve fiber can be deflected from its original path by an ultrastructural 'barrier,' such as is developing at the boundary of two substances of different consistency. Barriers crossing several nerve paths will cause the nerve fibers arriving on those paths to interlace and to form plexuses, and it is very likely that this condition is behind many a plexus formation in the organism. Formation of sub-epithelial plexuses, formation of plexuses on the boundary between different layers of the brain, and, possibly, formation of peripheral limb plexuses, could be caused in this way. Our experiments would, therefore, give an essentially true picture of plexus formation in the embryo and its mechanism. Although in the case of limb plexuses the concentration and displacement of the muscular material may account for the convergence of the segmental nerves (Eisler, '02; van Rynberk, '08), the mutual anasto-

mosing of the nerves will have to be explained on a different basis. Since the interconnections occur exactly at the girdle region, there is good reason to believe that it is the structural organization either of the girdle blastemas themselves or of their correspondingly oriented surroundings which causes nerve fibers to deviate at a sharp angle from their original course, until the peripherad movement of the limb mesenchyme drags them again into a centrifugal direction.

Furthermore, our experiments furnish a plausible explanation for the fact that nerves tend to grow toward developing organs, as if attracted by them. Detwiler ('22, '25), in a careful experimental study, has shown that the developing limb bud 'attracts' outgrowing nerves. It is significant to note that this apparent attraction on the limb nerves is not exhibited specifically by the limb bud but also by other organs in high developmental activity (nasal placode, Detwiler, '28; myotomes, Hoadley, '25). If we assume that the developing organ is a center of high chemical activity (Child), associated with some sort of dehydrating activity similar to the one exhibited by proliferating cultures in vitro, we may expect structures in the surrounding ground substances to develop which will converge upon the active center, comparable to the 'bridges' in our above experiments. The nerves happening to grow out at this time would then utilize the established pathways and so be led toward the differentiating center. The advantage of this interpretation of the facts over the chemotactic hypothesis will be discussed later.

Another significant phenomenon in the establishment of the embryonic nerve pattern, i.e., the formation of intracentral fiber tracts connecting two centers, finds its explanation easily on the basis of our experiments. Coghill ('29) has concluded from his exhaustive studies on the developing amphibian nervous system that "throughout the whole nervous system of *Amblystoma* the differentiation of neuroepithelial cells into nerve cells begins in localized centers. . . . Cells which differentiate in relative proximity to a center grow first toward that center, and cells which lie between two such

centers of acceleration orient their growth along lines between these centers. Under the operation of this law we find many of the important conduction systems of the brain laid down." This is exactly the situation that has been reproduced in our experiments in section I. 2, these experiments not only showing that fiber tracts really grow straight toward 'centers of accelerated growth,' but at the same time even revealing the mechanism to which this striking fact is due. This mechanism consists in the formation of a 'bridge' in the ground substance between two centers, such formation resulting from their dehydrating or similar activities. If two centers develop simultaneously, reciprocal growth of fiber tracts can result between them (fig. 18 a)—a fact for which chemotaxis can hardly account, since the influence of the center from which the fibers grow out would obviously be stronger than the influence of the farther distant, though equally active, center toward which they seem to tend. If two centers do not develop in exact synchronism, we will have to expect that the 'bridge' arising from their activity will be utilized only by the fibers from the younger center, as the fibers of the older center have already grown out previous to the formation of the 'bridge'; in this case fiber growth will be found to prevail in one direction (fig. 18 b). The fact that nerve fibers tend to grow into regions of high mitotic activity is made clear especially by Burr's ('32) experiments with transplanted placodes, where the aberrant nerves growing out from the transplanted placodes were found to grow into those parts of the wall of the brain where at the time the proliferative activity reached a maximum. Evidently our explanation holds true for these cases as well.

It might seem that our conception encounters difficulty if we were to explain the fact that fiber tracts of the same region are quite commonly found to cross each other. However, if the time order of their development is taken into account the difficulty disappears. The ground substance, especially in the earlier developmental stages, has to be considered as a very plastic material. Whenever a group of orienting factors

ceases to act the ground substance previously subjected to them will again be in a position to assume new structural arrangements under the influence of newly arising factors. Therefore, nerve fibers growing out at different times are likely to encounter the ground substance under patterns of quite different arrangement. The ground substance at the end of development can be compared, to a certain extent, to a photographic plate on which several pictures have been taken in succession. Accordingly, nerve patterns developed at different times in the same area cannot be expected to be congruous. Another possibility for the development of crossing nerve patterns, especially of such crossing at right angles, lies in the fact that in many cases where internal tensions arise from the change in form of an organ, two systems of forces (orthogonal trajectories) will be established, their lines of force crossing each other at right angles; the spongy bones illustrate the situation.

Although our experiments clearly demonstrate that between two centers of relatively high activity connecting fiber tracts must and do develop, it would be erroneous to assume that under normal conditions the laying down of nerve tracts is exclusively determined by this factor. The activity of growing centers in establishing patterns in the ground substance has to compete with a large series of other factors that work on the same ground substance; among these, mechanical tensions arising from differential growth and from changes in the external form of the organ, particularly in folding, may play an important role, and in fact many of the architectural details in the organization of the spinal cord and the brain seem by their obviously trajectorial arrangement to indicate that mechanical factors have operated in their establishment. It is intended to make these problems the objective of a later paper.

Our general conclusion, then, is that the essential and immediate principle in nerve orientation is the mechanical structure of the ground substance. The ideas of His (1887) and Harrison ('10) fall in line with this conception provided they

are extended to include ultrastructures in the range of guiding agents. Also Dustin's ('10) principle of hodgegenesis in nerve regeneration should be mentioned in this connection. Not in conformity with our conception is Vanlair's (1882, 1885) idea of minimum mechanical resistance being the guiding principle. If this were the case, nerve fibers should be expected to grow most easily in a liquid medium, where Harrison has shown almost no growth to occur ('12). Similarly we should expect that a nerve fiber coming from a firmer into a softer medium would continue to grow therein. Our experiments on plexus formation, however, have shown that the nerve fiber in such case tends to remain in the firmer medium owing to its particular ultrastructure. Therefore, the idea that the nerve fibers always choose as their pathways the lines of minimum mechanical resistance is not consistent with the facts. It is not a negative principle—absence of resistance—that guides the nerve fiber, but a positive principle—the presence of guiding structures; incidentally, these latter may be ways of minimum resistance. The fact that many peripheral nerves are found in close association with the main blood vessels, even under aberrant experimental conditions (Hamburger, '29) need not be interpreted as if blood vessels were the guiding structures for the outgrowing nerves but should rather be considered as an expression of the fact that both the primordial blood vessels and the nerve fibers are governed by a common agent, namely, the ultrastructure of the peripheral ground substance.

Still there exists an occurrence in development that can hardly be reconciled with the idea that nerve fibers depend upon underlying structures with absolute necessity. We see nerves perforating membranes, apparently working their way through microscopic structures that manifestly exist, as is the case for instance in the breaking of nerve roots through the external limiting membrane of the spinal cord. This phenomenon seems to imply an ability of the nerve fiber to destroy existing mechanical structures. On some rare occasions in our tissue culture experiments nerve fibers were observed

which seemed to have remained unaffected by the mechanical structure of the medium. A proteolytic activity of the growing tip of such fiber could account for that observation; at any rate, if such a proteolytic capacity really exists, it certainly is only temporarily exhibited under peculiar and exceptional conditions. Otherwise nerve fibers of this sort would have been observed more frequently. On the other hand, it has not even yet been proved that perforation of the limiting membrane is effected by the nerve itself. We shall have to keep in mind that the perforation might occur under some other influence, even without the participation of the nerve fiber. That such possibility exists is clearly illustrated by the perforation of the so-called operculum in tadpoles. The fore leg, after having developed in the gill cavity ruptures the overlying skin (operculum). One would suppose that the rupture is caused by the activity of the limb itself. However, Braus ('06) has found that the perforation of the operculum occurs at a normal time and at the typical location even if no limb is present, limb development being suppressed by an early removal of the limb blastema. Extensive confirmation of this fact can be found in the later work of Helff ('26). We see that perforations can occur without the supposedly perforating body playing any active part in it. Hence, the possibility should be at least considered that in perforation of nerve roots through the limiting membrane the situation might be similar.

On the whole there seems to be no unsurmountable difficulty that would prevent us from generalizing our structural hypothesis to the development of the nervous system in the normal embryo.

It remains to be considered whether besides the mechanical factor there is any place left for the assumption of chemical or electrical agents orienting in a direct way the outgrowing nerve fiber. Let us consider the chemical factors first. Concerning them, our tissue culture experiments gave a clear answer. No chemical substance was found which was capable of attracting nerve fibers *in vitro*; even the degenerating

nerve proved ineffective, though according to Cajal ('08) and Forssman (1898, '00), it could be assumed to be the source of most powerful neurotropic substances. The only case met with in our tissue cultures which formerly would certainly have been claimed as evidence for chemical attraction is the formation of fiber tracts connecting two cultures (section I. 2); but it is exactly this case which shows in the most convincing way that the orientation was not produced by direct chemical attraction. The fact that only a small fraction of the edge of each culture is involved in the effect, the fact that the mechanical structure of the 'bridge' can be seen before the fibers grow over it, and the further fact that pure brain cultures do not give the effect at all, constitute unequivocal evidence against the assumption of any direct chemotactic mechanism.

Of course, all of this evidence pertains to the conditions in tissue culture only, and the question will be raised whether in the organism there might not be chemotactic influences at work. In this connection the experiments of Cajal and Forssman on nerve regeneration will be cited as an apparently positive answer. However, a critical analysis of their brilliant experiments reveals that most of the facts can be explained as well on a structural as on a purely chemical basis, even in cases where it at first seems almost inevitable to assume selective chemical attractions. Already Dustin ('10), after failing to obtain chemotactic orientation effects in his experiments on nerve regeneration, has criticized Forssman's conclusions in the same sense, as we attempt to do here. He, too, was aware of the fact that the supposed 'chemotactic' effects could as well be interpreted as mechanical effects, due to cicatrization. In fact, Dustin succeeded in explaining four out of five of the principal experiments of Forssman on a structural basis; the fifth case, however, eluded him. This case was the following (Forssman, '20):

Forssman split the cut end of a nerve longitudinally and over each half he placed a tube, one containing crushed brain and the other crushed spleen or liver. Regeneration of nerve fibers took place from both halves but the fibers were found

to have grown considerably farther in the brain tube than in the spleen tube; this was, of course, explained as a specific chemical stimulation exerted by the brain tissue. Fortunately Forssman himself furnishes a clue for the structural explanation by stating: "Die Organisation der mit Milz und Leber gefüllten Röhrchen ist wenig nach oben fortgeschritten im Vergleich mit der Organisation der mit Gehirn gefüllten Röhrchen" (p. 429). Apparently what happened was that the proteolytic activity of the spleen fragments had prevented the content of the spleen tubes from 'organizing,' which means solidifying and developing definite structures, while in the brain tubes such structuration has occurred; it is, then, the structure present in the brain tubes rather than any specific chemical attraction that enabled the nerve fibers to grow into the brain tubes, whereas the lack of structures in the spleen tubes explains why nerve fibers made but slight progress therein. Herewith, we supplement Dustin's criticisms. Many more examples could be cited where the structural basis of supposedly chemical effects is equally clear. The structural effects have formerly simply been overlooked. This is not surprising since the ultrastructural viewpoint was not yet conceived, and when talking of structures one always had in mind threads, walls and similar coarse formations which were, of course, insufficient to account for nerve orientation.

In addition to the fact that real chemotaxis has not yet been proved to work on nerves, it may be mentioned that it is even unlikely that it could work: In case diffusion from the chemical source occurred in comparatively stiff gels, as for instance the interior of the organisms, it would be so slow and hampered that it would not reach any distant area. If diffusion took place in an almost liquid system, the local currents in that system would distort the main diffusion gradient to such an extent as to abolish it almost entirely. Finally, even if the possibility of direct chemical orientation should exist, it would be difficult to prove it, since in most cases localized chemical activity, if occurring in colloids, such as are found in the organism, will be associated with the formation of correspondingly oriented structures.

If we are skeptical concerning the possibility of a direct chemical attraction of nerve fibers without the interaction of leading mechanical structures, this should not mean that chemical activity is not operating at all. On the contrary, we have shown how chemical activity participates essentially, sometimes determinatively, in the establishment of mechanical structures, and hence, though only secondarily, in the orientation of the nerve fiber. Furthermore, it must be admitted that the traffic of chemical substances along the mechanical structures can affect the nerves growing thereon. Although the mere fact that the transportation of liquids is facilitated along the oriented structures can account for the enhancement of growth in these directions (see above), it is possible that specific chemical agents carried along that system of traffic lines exhibit certain stimulative effects. If a nerve fiber encounters a cross path, as in our plexus experiments (section I. 3), it can enter the cross path either by a left or a right turn. Which way will it go? In our experiments the decision seemed to depend entirely on the angle of approach: The nerve fibers always deflect obtusely, never at an acute angle. It may be that in the organism under similar conditions agents running down the new path in a constant direction may help to determine in which direction the fiber will turn. In many cases, however, the fibers branch in entering the cross path and grow in both directions (development of longitudinal tracts in the spinal cord).

Summarizing we may say that while the indirect chemical influences on nerve orientation are very essential and vigorous, no conclusive evidence is at hand in favor of direct chemical orientation effects operating without the intervention of the structural factor.

Finally a few words must be said about the electric current. As far as our experiments go, the electric current is completely ineffective in orientation of nerve fibers. Ingvar's ('20) results which seemed to prove the contrary, will have to be explained by his introducing mechanical orientation effects without being aware of it. As far as the embryo is concerned,

the absence of electric factors in orienting nerve fiber growth is already proved beyond doubt by the mere observation that the same pathway is frequently utilized at the same time by two nerve fibers growing in opposite directions. This is known to occur in the development of reciprocal fiber tracts in the central nervous system and has been observed to occur in the development of peripheral nerve fibers as well (Speidel, '33). Again, the only way in which electric agents might possibly influence the orientation of nerve fibers is by an orienting effect on the ground substance such as could be imagined to arise both from direct polarization and from cataphoresis. But even effects of that sort have not been obtained in the experiments so far. Therefore, it can be regarded as almost certain that electric currents are not among the agents orienting the course of nerve fibers. They might, however, play a role in polarization and migration of neuroblasts.

Carrying the evaluation of our experimental results further than has been attempted in the preceding discussion would mean to overload again with speculation a foundation of facts which is still narrow. Our task has been to demonstrate that 'ultrastructural organization' may be the mechanism of nerve fiber orientation in the organism as well as in vitro. Whether it actually is the mechanism remains to be seen. At present, all we can state in its favor is: that it implies a concrete, intelligible and reproducible mechanism, that the range of its applicability is apparently wider and more universal than that of the chemical, electrical and mechanical hypotheses advanced in the past, that it enables us to explain a great variety of facts on a common and uniform basis, and that it is, at any rate, not contradictory to what is known about development, regeneration and pathology of the nervous system. In order to avoid misinterpretation, utmost stress must be laid on the statement that the principle of 'ultrastructural organization' does not in any sense antagonize or preclude the assumption of chemical, electrical, radiant, or even undiscovered agents to be among the factors deter-

mining the spatial arrangement of nerves; on the contrary, it allows for the play of all these agents, indicating beyond the mere acknowledgment of their intervention the manner how they intervene and operate: In dealing with the nerve fiber, they all use the ground substance with its capacity for 'ultrastructural organization' as their common executive intermediary.

Thus far we have considered only the orientation of the nerve fiber on its course and the bearing of our experiments on that particular problem. This, however, represents only part of the complex and intricate problem of nerve patterns. Other component processes in nerve pattern formation are: polarization of the neuroblasts; migration and orientation of the neuroblasts; branching of the nerve fibers at given points; association of nerve fibers into bundles; branching of nerve trunks; final connection of nerve fibers with other nerve cells and with peripheral organs. Furthermore, there may be a controlling organismic factor in play, coordinating in space, time and quantity all the component activities.⁶ Each one of the component activities mentioned must be subjected to a separate test, if we are to find out about their respective mechanisms, and whether a common mechanism might not exist for several of the components. Little as our knowledge about the orientation of the nerve fiber is, the actual information about these other phenomena of nerve development is still less. No special attention has been given to them in the present work, either. But there is some casual information concerning them contained in our experiments, and this may be briefly discussed:

*The existence of such 'supervising' factors determining and coordinating the activities of their subordinated parts in development is clearly evidenced by modern experimental embryology. In an attempt to describe in an adequate way the functioning of such determining and coordinating dynamics in developing systems, the notion of 'embryonic fields' has been conceived by Gurwitsch and elaborated into some detail with special reference to regeneration by myself, without any commitment as to their nature (compare Weiss, '30). The fundamental properties of these 'embryonic fields,' as deduced from their manifestations in embryogenesis and regeneration, point unmistakably to the improbability of their being identical with ordinary electric fields as recently suggested by Burr ('32).

Migration of neuroblasts. Mobilization of neuroblasts from the explanted fragment has been observed, but only in exceptional cases; in liquefying cultures where the occurrence is more frequent (compare Olivo, '27, '28) the passive displacement of the cells interferes with the display of their active orientation with the effect that no definite conclusions can be reached. Incidentally, in a few cultures which were exposed to the electric current, some neuroblasts became detached from the central fragment, apparently in the direction of the current. The phenomenon was not marked enough to allow any inferences although it makes us think of Kappers' ('17) hypothesis, according to which electric potential differences cause polar migrations of neuroblasts. It is equally possible that in our experiments the electric current has simply induced the cells opposite the electrodes to liquefy the medium, the retraction of which would then, secondarily, have mobilized the cells.

Branching of nerve fibers. This was frequently observed, but the conditions under which it occurred did not show any perceptible regularity except that branching failed to take place while a nerve fiber was traveling in a region of strictly parallel organization. Generally speaking, branching is the more frequent, the less regularity there is in the structures of the medium. The experiences made with nerves regenerating in the organism lead to the same conclusion, as excessive branching of regenerating nerve fibers is always observed in the location of least regularity, i.e., in the scar (Cajal, '08). Less regularity means more mechanical obstruction encountered by the nerve fiber on its way. Mechanical obstruction, therefore, not only deflects a nerve fiber but sometimes seems able to cause it to divide. Observations of this sort have been made on the living nerve fiber in the organism by Speidel ('33), and the haphazard branching of nerve fibers during regeneration is certainly due to the same fact. On the other hand, it seems impossible to consider mechanical resistance encountered by the nerve fiber as the typical or even as the essential factor effecting branching. Never could such an

assumption account, for instance, for the formation of collaterals which do not branch off the growing tip, but arise at some later time as a lateral outgrowth of the stem of the fiber. To explain such cases, some local influence must be assumed which, by altering the local condition of the passing nerve fiber, causes this latter to produce a branch at the given point. A weak spot, so to speak, must be created on the nerve fiber by some local agent. Only secondarily may 'ultrastructural organization' be involved in this process, in so far as the local flow and diffusion of liquids is directed by the structures of the ground-substance so as to expose to all sorts of stimuli any particular point of a nerve fiber against which an oriented structure is directed. An experimental analysis of this problem in tissue culture seems entirely feasible.

Branching of nerve bundles. In this phenomenon in which ramification of single axons is not necessarily involved, the mechanical conditions at the point of branching may be decisive, as was already assumed by His (1887). Examples of this have been occasionally observed in the tissue culture experiments also. In the embryo, however, the particular branching pattern will depend largely on the growth, mutual shifting, and change in form which the peripheral parts undergo after their first nervous connections have already been established; apparently the later concentration of previously separated fiber bundles into thicker 'stems' is as real an occurrence during embryonic development as is the separation of bundles out of stems.

Association of nerve fibers. Why do nerve fibers under certain circumstances tend to aggregate and to form nerves, while they fail to do so under other circumstances? It seems that with respect to this problem, too, some information can be obtained from our tissue culture experiments. As a general experience it can be stated that in a clot of normal composition the fibers show little, if any, inclination to associate (compare fig. 9). Association takes place, however, if the amount of liquid in the medium surpasses a certain threshold. In areas of liquefaction, on the free surface of the clot, in the

space between clot and cover slip, in any location where liquids are accumulated, numbers of fibers can be observed sticking together and forming bundles, sometimes of considerable size. The matter is clearly illustrated by figure 10. There we see those fibers which are situated in the slightly liquefied surface of the clot, the same ones that farther peripherally enter into the plexus, take their course not as single fibers, but united into several strong bundles. These radial bundles, appearing as thick lines in the picture, contrast sharply with the thin lines in the background which represent the mass of single fibers failing to associate as they grow in the more solid interior of the clot. Translated into the terms of embryogenesis, this observation would mean that nerve fibers will be found to collect into bundles wherever their paths lead through regions abundant in liquids (interspaces between organs, surfaces of membranes, meshes of loose connective tissue), that the bundles will dissociate again into single fibers, as soon as they have entered tissues of appreciable consistence. True though this may be in many cases, it is not advisable to lay stress on any such analogy in advance of a thorough investigation of the situation in the embryo itself.

Specificity of peripheral connection between nerve fiber and end organ. We are still completely in the dark as to the factor that under normal conditions makes a motor fiber connect with a motor end-organ, and a sensory fiber with a sensory periphery. Generally the idea is held that some specific affinity exists between any peripheral part and the special sort of nerve fiber that is intended to innervate it; it is assumed that the end-organ, by virtue of its specificity, either can attract the appropriate nerve fibers from a distance or can select them out of a non-assorted group when arriving in the vicinity. The only thing we can consider as settled, at any rate, is that if particular tissues really possess such power to select their respective nerves, they certainly possess it only in the embryonic stage and lose it after that. For, it has been demonstrated beyond doubt by Boeke ('21)

that in the adult organism sensory fibers can be brought to establish connections with a motor organ, and motor fibers with a sensory periphery. I myself have shown that connections, even functional, can be effected between motor fibers and the gray matter of the spinal cord (P. Weiss, '32), and between the dorsal root fibers of spinal ganglia and muscles.⁷ All these experiments prove that, at least in the adult organism, no selectivity exists to control the formation of peripheral nervous connections. Still the possibility remains that such selectivity might exist during embryogenesis. A first attempt to approach the problem by the method of tissue culture was made by Harrison ('10), but no decisive results were obtained. I repeated the experiments several times; but the results were again entirely negative: A spinal ganglion to produce sensory fibers, a piece of spinal cord to produce motor fibers, a fragment of skin and a muscle, all taken from the same embryo, were included in a common clot and arranged in a square, the two nerve cultures in opposite corners, so that the distance to grow for the motor or sensory fibers was the same both to skin and muscle. In no case was there any preferential ingrowth noticeable of motor fibers into muscle or of sensory fibers into skin, both sorts of fibers growing as well and as abundantly into both sorts of tissues. One could object that the tissues or the nerves might have lost their specific abilities under the abnormal conditions of tissue culture; although other types of cells have been proved to retain their specific abilities and qualities even after prolonged cultivation in vitro (A. Fischer, '30), the objection has not been invalidated in the present case of skin, muscle and nerve. Therefore, it may well be that the negative results in our experiments are meaningless. Should further experimentation on the embryo itself disprove the assumption of selectivity, then the time order according to which different types of nerves develop at slightly different times and rates, might provide an explanation; if, for instance, at the time the motor fibers reach the periphery only the muscles are in

⁷ These experiments are to be published in *The Journal of Comparative Neurology*.

a proper state to accept innervation, while the sensory areas are not ready for innervation, and the sensory areas are ready at a later moment when the sensory fibers arrive, specificity of peripheral connections would be accounted for. Experiments on this matter are under way.

SUMMARY

1. An attempt was made to determine the factors involved in the orientation of nerve fibers by exposing in-vitro cultures of nerve-producing neuroblasts to a series of mechanical, chemical and electrical agents.

2. By stroking or stretching the clot of blood plasma serving as culture medium in a constant direction, an oriented ultrastructure (micellar orientation and aggregation) can be evoked therein. The nerves growing in such organized clot follow distinctly the pre-established orientation, utilizing the ultrastructures as guiding tracks. The pattern of nerve fibers is, therefore, merely a reproduction of the ultrastructural pattern existing in the surrounding ground-substance at the time of the outgrowth of the fibers.

3. Two outstanding phenomena in the development of nerve patterns could be imitated in vitro and analyzed as to their mechanism: the formation of plexuses, and the establishment of connecting fiber tracts between localized centers.

4. If an ultrastructure is evoked in the medium such as to cross the general direction of outgrowth, the nerve fibers arriving at the cross structure (ultrastructural 'barrier') are deflected from the original paths, forming a plexus along the 'barrier.' It is argued that, similarly, transversal structures lying across the way of the growing nerve fibers (layers of the brain, girdle blastemas) are responsible for the formation of nerve plexuses in the embryo.

5. The formation of connecting fiber tracts is strikingly demonstrated when two spinal ganglia are cultivated in a thin plasma membrane at some distance from each other: A narrow streak of cells and nerve fibers grows then in a perfectly straight direction along the line connecting the two cultures.

The assumption that the phenomenon is due to 'chemotactic' attraction could be decisively disproved. The real mechanism is as follows: The proliferative activity of the cultures entails dehydration of the surrounding medium. The dehydration causes contraction. Contraction proceeding from two centers establishes tensions which, in the space between the two centers, reach a maximum of intensity and are oriented in the direction of the connecting line. These oriented tensions evoke a correspondingly oriented pattern of ultrastructures in the medium, an ultrastructural 'bridge,' leading from one center to the other. The outgrowing nerve fibers simply follow the 'bridge.' On the basis of these results the fact is explained that in embryogenesis intracentral fiber tracts develop between centers of accelerated growth (Coghill) and that peripheral nerves are 'attracted' by growing organs (Detwiler).

6. Outgrowing nerve fibers were exposed to localized sources of chemical diffusion; to brain extract, muscle extract, liver extract, and degenerating peripheral nerve. In no case was there any 'chemotactic' orientation effect observed with respect to these agents. Many effects that formerly were supposed to be 'chemotactic' in nature can be explained as structural effects resulting from a participation of chemical activities in the formation of the leading ultrastructures in the ground-substances.

7. Outgrowing nerve fibers were in various ways exposed either to the electric current or to the electromagnetic field around a conductor. No orientation effect whatsoever was obtained in either case.

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